

Khachiyan's Inscribed-Ellipsoid Cutting Inequality: Mathematical Specification and Complete Proof

0. Definitions

Fix an integer $n \geq 1$. Work in $\mathbb{R}^n$ with its standard inner product and Euclidean norm. Let vol_n denote Lebesgue volume, $B_n = \{x : \|x\| \leq 1\}$, and $\kappa_n = \text{vol}_n(B_n) \in (0, \infty)$.

A convex body is a compact convex set with nonempty interior. For a real symmetric positive definite matrix A and a vector $a \in \mathbb{R}^n$, define

$$\mathcal{E}(a, A) = a + AB_n.$$

Write $X \preceq Y$ when $Y - X$ is positive semidefinite. If $X = U \text{diag}(\lambda_i)U^\top \succ 0$, define, for $p \in \mathbb{R}$,

$$X^p = U \text{diag}(\lambda_i^p)U^\top.$$

For positive semidefinite matrices, use the same definition when $p > 0$, with the convention $0^p = 0$.

For a convex body K , define

$$w(K) = \max\{\text{vol}_n(\mathcal{E}(a, A)) : A = A^\top \succ 0, \mathcal{E}(a, A) \subset K\}.$$

The maximum exists and the maximizing ellipsoid is unique, as proved in G03. Denote this ellipsoid by $J(K)$ and its center by $c(K)$. Define the exact constant

$$r_* = \frac{e^{1/2}}{2} = \frac{\sqrt{e}}{2}.$$

1. Main statements

T01: Geometric upper bound. For every $n \geq 1$, convex body $K \subset \mathbb{R}^n$, and vector $p \neq 0$, let

$$H = \{x : p^\top(x - c(K)) \geq 0\}.$$

Then $K \cap H$ is a convex body and

$$w(K \cap H) \leq r_* w(K).$$

Replacing p by $-p$ gives the same conclusion for the other side.

T02: Uniform optimality. For every real number $r < r_*$, there exist an integer $n \geq 2$, a convex body $K \subset \mathbb{R}^n$, and a closed halfspace H whose boundary passes through $c(K)$ such that

$$w(K \cap H) > r w(K).$$

2. Geometry and maximality

G01: Volume, support functions, and affine transformations

For $A \succ 0$,

$$\text{vol}_n(\mathcal{E}(a, A)) = \kappa_n \det A, \quad h_{\mathcal{E}(a, A)}(v) = v^\top a + \|Av\|.$$

Here h denotes the support function. The volume identity follows from translation invariance of Lebesgue measure and its determinant scaling under invertible linear maps. The support identity follows from

$$\max_{\|y\| \leq 1} v^\top (a + Ay) = v^\top a + \max_{\|y\| \leq 1} (Av)^\top y$$

and the Cauchy--Schwarz inequality. If $Av \neq 0$, the maximum is attained at $y = Av/\|Av\|$.

Every invertible linear image CB_n equals AB_n , where $A = (CC^\top)^{1/2} \succ 0$: the matrix $A^{-1}C$ is orthogonal. Consequently, an invertible affine transformation gives a bijection on nondegenerate ellipsoids and scales all volumes by the same factor. Maximality among inscribed ellipsoids and volume ratios are therefore preserved.

For a nonempty compact convex set K ,

$$K = \{x : \forall v, v^\top x \leq h_K(v)\}.$$

To prove the reverse inclusion, suppose $x \notin K$. Choose $y \in K$ minimizing the distance to x , and put $v = x - y \neq 0$. For $z \in K$, the right derivative at $t = 0$ of the squared distance along $y + t(z - y)$ gives $v^\top(z - y) \leq 0$. Hence $h_K(v) \leq v^\top y < v^\top x$, contradicting the assumed support inequalities.

G02: Intermediate ellipsoid

If a convex body K contains B_n and $\mathcal{E}(b, S)$, with $S \succ 0$, then

$$\mathcal{E}\left(\frac{b}{2}, \left(S + \frac{1}{4}bb^\top\right)^{1/2}\right) \subset K.$$

Proof. For $v \neq 0$, write $h = h_K(v) > 0$ and $s = v^\top b$. By G01 and Cauchy--Schwarz,

$$\|v\| \leq h, \quad \|Sv\| \leq h - s, \quad v^\top Sv \leq h(h - s).$$

It follows that

$$v^\top (S + bb^\top/4)v \leq (h - s/2)^2.$$

Since $s \leq h$, we have $h - s/2 > 0$. The support function of the proposed ellipsoid is therefore at most

$$s/2 + \sqrt{v^\top (S + bb^\top/4)v} \leq h.$$

The inequality also holds for $v = 0$. G01 now gives the required containment.

G03: Existence and uniqueness of the maximum-volume inscribed ellipsoid

Temporarily allow positive semidefinite shape matrices $A \succeq 0$. Feasibility implies $a \in K$ and $a \pm Au \in K$, so $2\|Au\| \leq \text{diam } K$ for every unit vector u . Thus the feasible parameter set is bounded.

Containment and positive semidefiniteness are preserved under limits, so this set is closed and hence compact. The continuous function $\det A$ attains a maximum. Since K contains a ball of positive radius, the maximum is positive, and every maximizing matrix is positive definite.

The feasible parameter set is convex: for $y \in B_n$,

$$(1 - t)(a_0 + A_0 y) + t(a_1 + A_1 y) \in K \quad (0 \leq t \leq 1).$$

Suppose (a_0, A_0) and (a_1, A_1) are two maximizing parameters. Let $C = A_0^{-1/2} A_1 A_0^{-1/2} \succ 0$, with eigenvalues $\mu_i > 0$. The scalar inequality $(1 + \mu_i)/2 \geq \sqrt{\mu_i}$ gives

$$\det \frac{A_0 + A_1}{2} = \det A_0 \prod_i \frac{1 + \mu_i}{2} \geq \det A_0 \prod_i \sqrt{\mu_i} = \sqrt{\det A_0 \det A_1}.$$

If $A_0 \neq A_1$, then $C \neq I$, so at least one $\mu_i \neq 1$ and the inequality is strict. The midpoint would then be feasible with a larger determinant, contradicting maximality. Thus $A_0 = A_1$.

If the centers differ, affine normalization gives two maximizing ellipsoids B_n and $b + B_n$, with $b \neq 0$. G02 produces a feasible shape matrix $(I + bb^\top/4)^{1/2}$ whose determinant is $\sqrt{1 + \|b\|^2/4} > 1$, again contradicting maximality.

G04: Normalization and center exclusion

The affine map $x \mapsto S_*^{-1}(x - a_*)$ sends $J(K) = a_* + S_* B_n$ to B_n . For a candidate ellipsoid $E = \mathcal{E}(a, A)$ in the normalized coordinates, its volume relative to the unit ball is $\det A$, and

$$0 \notin \text{int } E \iff \|A^{-1}a\| \geq 1.$$

Indeed, the unique solution of $0 = a + Ay$ is $y = -A^{-1}a$. A nondegenerate ellipsoid contained in the halfspace $p^\top x \geq 0$ cannot contain 0 in its interior.

Conversely, if $\rho = \|A^{-1}a\| \geq 1$, choose $v = A^{-2}a \neq 0$. Then

$$\min_{x \in E} v^\top x = v^\top a - \|Av\| = \rho^2 - \rho \geq 0.$$

Thus the center-exclusion formulation is equivalent to the formulation using halfspaces through the original center.

G05: Necessary trace condition

If $J(K) = B_n$ and $\mathcal{E}(b, S) \subset K$, with $S \succ 0$, then $\text{tr } S \leq n$.

Proof. For $y \in B_n$, convexity gives $(1 - t)y + t(b + Sy) \in K$. Consequently,

$$tb + ((1 - t)I + tS)B_n \subset K \quad (0 \leq t \leq 1).$$

Therefore $\log \det((1 - t)I + tS) \leq 0$, with equality at $t = 0$. Taking the right derivative yields $\text{tr}(S - I) \leq 0$. This derivative can also be computed directly by diagonalizing S and differentiating $\sum_i \log(1 + t(\lambda_i - 1))$.

3. Matrix analysis

M01: Spectral calculations

For $A \succ 0$, write its eigenvalues as $\lambda_i > 0$ and put $\alpha = \min_i \lambda_i > 0$. Then

$$\det A = \prod_i \lambda_i, \quad \text{tr}(A^p) = \sum_i \lambda_i^p, \quad A \preceq A^2/\alpha.$$

For $p > 0$ and a unit vector u ,

$$u^\top A^{2p} u \geq \alpha^{2p}.$$

These statements follow coordinatewise in an orthonormal eigenbasis of A . If $X \preceq Y$, then $\text{tr } X \leq \text{tr } Y$, since the diagonal entries of the positive semidefinite matrix $Y - X$ are nonnegative.

M02: Order properties of inverses and fractional powers

If $0 \prec X \preceq Y$, then $Y^{-1} \preceq X^{-1}$. Indeed, write $Y = X^{1/2} C X^{1/2}$. Then $C \succeq I$, hence $C^{-1} \preceq I$, and a congruence transformation gives the conclusion.

For $0 < p < 1$, define

$$c_p = \left(\int_0^\infty \frac{s^{p-1}}{1+s} ds \right)^{-1} > 0.$$

The integral converges and is positive because $0 < p < 1$. The substitution $s = xt$ for $x > 0$, followed by orthogonal diagonalization, gives

$$X^p = c_p \int_0^\infty s^{p-1} X (X + sI)^{-1} ds. \quad (\text{M02.1})$$

Since $X(X + sI)^{-1} = I - s(X + sI)^{-1}$, inverse antitonicity shows that the matrix-valued integrand is increasing in X . Therefore

$$X^p \preceq Y^p \quad (0 < p < 1).$$

Combining this with $(X^{1-p})^{-1} = X^{p-1}$ gives

$$X^{p-1} \succeq Y^{p-1} \quad (0 < p < 1).$$

In particular, the square root preserves matrix order.

M03: Trace differentiation along self-adjoint directions

Let $X \succ 0$, $H = H^\top$, and $0 < p < 1$. At $t = 0$,

$$\frac{d}{dt} \text{tr}((X + tH)^p) = p \text{tr}(X^{p-1} H).$$

Proof. The matrix $X + tH$ remains positive definite in a neighborhood of zero. Matrix inversion is differentiable on the invertible matrices, as its entries are rational functions expressed using the adjugate and determinant. Put $R_s = (X + sI)^{-1}$. Taking the trace in (M02.1) and differentiating yields

$$D[\text{tr}(X^p)](H) = c_p \int_0^\infty s^p \text{tr}(R_s^2 H) ds.$$

Differentiation under the integral is justified as follows. In a positive definite neighborhood of X within the self-adjoint matrices, the norm of the differentiated integrand is bounded by Cs^p near zero and by Cs^{p-2} at infinity. Both bounds are integrable. In an eigenbasis of X , integration by parts gives

$$c_p \int_0^\infty \frac{s^p}{(\lambda + s)^2} ds = p\lambda^{p-1},$$

because the boundary term $s^p/(\lambda + s)$ vanishes at both endpoints. This proves the formula. The argument requires H to be self-adjoint but does not require $XH = HX$.

M04: Quantitative rank-one trace estimate

Let $A \succ 0$, $\alpha = \lambda_{\min}(A)$, $\|u\| = 1$, and $b = Au$. If $M \succ 0$ and $m > 0$ satisfy

$$M \preceq \frac{m}{\alpha^2} A^2,$$

then, for every $0 < p < 1$ and $t \geq 0$,

$$\operatorname{tr}((M + tbb^\top)^p) - \operatorname{tr}(M^p) \geq (m + t\alpha^2)^p - m^p. \quad (\text{M04.1})$$

Proof. Since $uu^\top \preceq I$, we have $bb^\top \preceq A^2$. Thus, for $s \geq 0$,

$$M + sbb^\top \preceq \frac{m + s\alpha^2}{\alpha^2} A^2.$$

Apply the negative-power antitonicity from M02 and evaluate the resulting quadratic form on $b = Au$. This gives

$$\begin{aligned} b^\top (M + sbb^\top)^{p-1} b &\geq \left(\frac{m + s\alpha^2}{\alpha^2} \right)^{p-1} u^\top A^{2p} u \\ &\geq \alpha^2 (m + s\alpha^2)^{p-1}. \end{aligned}$$

By M03, multiplying the left side by p gives the derivative of $\operatorname{tr}((M + sbb^\top)^p)$. Integration over $[0, t]$ proves (M04.1).

If u is a unit eigenvector of A corresponding to α and $M = (m/\alpha^2)A^2$, the perturbation changes only the eigenvalue m in that direction. Equality therefore holds in (M04.1).

M05: Diminishing trace increments

If $0 \prec X \preceq Y$, $H \succeq 0$, and $0 < p < 1$, then

$$\operatorname{tr}((X + H)^p) - \operatorname{tr}(X^p) \geq \operatorname{tr}((Y + H)^p) - \operatorname{tr}(Y^p).$$

Put $Z = Y - X \succeq 0$ and apply M03 to

$$g(t) = \operatorname{tr}((X + tZ + H)^p) - \operatorname{tr}((X + tZ)^p)$$

By M02, the derivative is p times the trace of a negative semidefinite matrix multiplied by Z , and is therefore nonpositive. Indeed, if $W \preceq 0$ and $Z \succeq 0$, then $\operatorname{tr}(WZ) = \operatorname{tr}(Z^{1/2}WZ^{1/2}) \leq 0$. Integration gives the conclusion.

Equivalently, for $U, V, W \succeq 0$,

$$\operatorname{tr}((U + V + W)^p) + \operatorname{tr}(W^p) \leq \operatorname{tr}((U + W)^p) + \operatorname{tr}((V + W)^p).$$

First take $X = W$, $Y = V + W$, and $H = U$ when $W \succ 0$. Then replace W by $W + \varepsilon I$ and let $\varepsilon \downarrow 0$, using continuity.

4. Two spectral constraints and the sharp scalar bound

Throughout this section, assume $J(K) = B_n$, $\mathcal{E}(a, A) \subset K$, $A \succ 0$, and $\rho = \|A^{-1}a\| \geq 1$. Write $\alpha = \lambda_{\min}(A)$.

R01: Two iterations

By G02 and G05,

$$T_1 = (A + aa^\top/4)^{1/2}, \quad T_2 = (T_1 + aa^\top/16)^{1/2}, \quad \text{tr } T_1, \text{tr } T_2 \leq n.$$

The corresponding centers are $a/2$ and $a/4$, respectively. Set $u = \rho^{-1}A^{-1}a$ and $b = Au = \rho^{-1}a$, and define

$$M_1 = (A + bb^\top/4)^{1/2}, \quad M_2 = (M_1 + bb^\top/16)^{1/2}.$$

Since $bb^\top \preceq aa^\top$, M02 gives $M_j \preceq T_j$, hence $\text{tr } M_j \leq n$.

Define

$$f_1(\alpha) = \sqrt{\alpha + \alpha^2/4}, \quad f_2(\alpha) = \sqrt{f_1(\alpha) + \alpha^2/16}.$$

Apply M04 with $(M, m, p, t) = (A, \alpha, 1/2, 1/4)$. Since $A \preceq A^2/\alpha$, this yields

$$\text{tr } M_1 \geq \text{tr}(A^{1/2}) + f_1(\alpha) - \alpha^{1/2}.$$

Moreover,

$$M_1 \preceq (A^2/\alpha + A^2/4)^{1/2} = \frac{f_1(\alpha)}{\alpha} A \preceq \frac{f_1(\alpha)}{\alpha^2} A^2.$$

Applying M04 with $(M, m, p, t) = (A, \alpha, 1/4, 1/4)$ and $(M_1, f_1(\alpha), 1/2, 1/16)$, respectively, gives

$$\text{tr}(M_1^{1/2}) - \text{tr}(A^{1/4}) \geq f_1(\alpha)^{1/2} - \alpha^{1/4},$$

$$\text{tr } M_2 - \text{tr}(M_1^{1/2}) \geq f_2(\alpha) - f_1(\alpha)^{1/2}.$$

Adding these inequalities cancels the intermediate terms, so

$$\text{tr } M_2 \geq \text{tr}(A^{1/4}) + f_2(\alpha) - \alpha^{1/4}.$$

Label one occurrence of the minimum eigenvalue as $\lambda_1 = \alpha$. M01 then gives

$$f_j(\alpha) + \sum_{i=2}^n \lambda_i^{1/2^j} \leq n, \quad j = 1, 2. \tag{R01.1}$$

R02: Determinant bounds

The inequality $\log z \leq z - 1$ for $z > 0$, together with (R01.1), yields

$$\begin{aligned} \log \det A &= \log \alpha + 2^j \sum_{i=2}^n \log(\lambda_i^{1/2^j}) \\ &\leq \log \alpha + 2^j(1 - f_j(\alpha)). \end{aligned}$$

Thus, on defining

$$R_j(\alpha) = \alpha \exp(2^j(1 - f_j(\alpha))),$$

we obtain $\det A \leq \min\{R_1(\alpha), R_2(\alpha)\}$. When $n = 1$, the sums are empty and the conclusion remains valid.

For $n \geq 2$, the arithmetic–geometric mean inequality also gives

$$\det A \leq \alpha \left(\frac{n - f_j(\alpha)}{n - 1} \right)^{2^j(n-1)}, \quad j = 1, 2. \quad (\text{R02.1})$$

The base is positive because (R01.1) contains $n - 1$ positive summands.

S01: Joint scalar maximum

Put $x = \log \alpha$ and $F_j(x) = \log R_j(e^x)$. The functions

$$\log f_1(e^x) = \frac{1}{2} \log(e^x + e^{2x}/4), \quad \log f_2(e^x) = \frac{1}{2} \log(e^{\log f_1(e^x)} + e^{2x}/16)$$

are convex. Indeed, $L(s, t) = \log(e^s + e^t)$ has positive partial derivatives and Hessian

$$q(1 - q) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \succeq 0, \quad q = \frac{e^s}{e^s + e^t}.$$

Composition with a convex function that is nondecreasing in each argument proves both asserted convexity statements. Every positive log-convex function is convex, so $F_j(x) = x + 2^j(1 - f_j(e^x))$ is concave for each j .

Let $x_0 = -\log 2$ and $C = 1/2 - \log 2$. Direct calculation gives

$$f_1(1/2) = 3/4, \quad f_2(1/2) = 7/8, \quad f'_1(1/2) = 5/6, \quad f'_2(1/2) = 43/84,$$

and hence

$$F_1(x_0) = F_2(x_0) = C, \quad F'_1(x_0) = 1/6, \quad F'_2(x_0) = -1/42.$$

The tangent bounds for concave functions imply

$$F_1(x) \leq C + \frac{x - x_0}{6}, \quad F_2(x) \leq C - \frac{x - x_0}{42}. \quad (\text{S01.1})$$

Use the first bound when $x \leq x_0$ and the second when $x \geq x_0$. This gives $\min(F_1, F_2) \leq C$, with equality at $x = x_0$. Therefore

$$\max_{\alpha > 0} \min\{R_1(\alpha), R_2(\alpha)\} = e^C = r_*.$$

Equivalently, the concave function $G = F_1/8 + 7F_2/8$ has derivative zero at x_0 , so $\min(F_1, F_2) \leq G \leq C$.

5. Proof of T01

By G04, normalize the original maximal ellipsoid to the unit ball and represent any nondegenerate ellipsoid inscribed in the retained side as $\mathcal{E}(a, A)$ satisfying the center-exclusion condition. R02 and S01 give $\det A \leq r_*$. Transforming back by G01 shows that its volume is at most $r_* w(K)$.

Since $c(K)$ lies in the interior of K , a hyperplane through this point intersects K on both sides in compact convex sets with nonempty interior. Apply the preceding estimate to the maximum-volume inscribed ellipsoid of $K \cap H$ to obtain T01.

6. Cones and uniform optimality

C01: Sufficient contact-point certificate

If $B_n \subset K$ and there are unit vectors u_i and positive weights $c_i > 0$ such that

$$K \subset \{x : u_i^\top x \leq 1\} \quad \text{for all } i, \quad \sum_i c_i u_i = 0, \quad \sum_i c_i u_i u_i^\top = I,$$

then $J(K) = B_n$.

Proof. Every $\mathcal{E}(a, A) \subset K$ satisfies $u_i^\top a + \|Au_i\| \leq 1$ and $u_i^\top Au_i \leq \|Au_i\|$. Taking weighted sums gives $\text{tr } A \leq \sum_i c_i = n$, so $\det A \leq (\text{tr } A/n)^n \leq 1$. The unit ball is feasible, and uniqueness follows from G03.

C02: Explicit cones

For $n \geq 2$, write $x = (t, y) \in \mathbb{R} \times \mathbb{R}^{n-1}$ and define

$$K_n = \{(t, y) : t \geq -1, \quad t + \sqrt{n^2 - 1} \|y\| \leq n\}.$$

This set is closed and convex. Its points satisfy $-1 \leq t \leq n$ and $\|y\| \leq (n+1)/\sqrt{n^2-1}$, so it is bounded and hence compact. Moreover, $B_n \subset K_n$: on the unit ball, $t \geq -1$, and Cauchy--Schwarz gives $t + \sqrt{n^2-1} \|y\| \leq n$. Thus K_n has nonempty interior.

Take the base normal $u_0 = -e_1$ and the side normals

$$u_v = \left(\frac{1}{n}, \sqrt{1 - \frac{1}{n^2}} v \right), \quad v \in \{\pm e_1, \dots, \pm e_{n-1}\} \subset \mathbb{R}^{n-1}.$$

Assign weight $c_0 = n/(n+1)$ to the base and weight $c_v = n^2/(2(n^2-1))$ to each side direction. The total side weight is $n^2/(n+1)$, so the first-coordinate first moment is zero and its second moment is 1. Transverse first moments and mixed second moments vanish by pairing opposite signs. Every transverse second moment is $2c_v(1 - 1/n^2) = 1$. All u_i are unit vectors and supporting directions of K_n . C01 therefore gives $J(K_n) = B_n$ and $c(K_n) = 0$.

Take $H_n = \{t \leq 0\}$ and

$$E_n = -\frac{1}{2}e_1 + \text{diag} \left(\frac{1}{2}, \sqrt{\frac{n}{n-1}}, \dots, \sqrt{\frac{n}{n-1}} \right) B_n.$$

Its first coordinate lies in $[-1, 0]$. For every unit transverse vector v , its support function in the corresponding side-normal direction is

$$-\frac{1}{2n} + \sqrt{\frac{1}{4n^2} + \frac{n}{n-1} \left(1 - \frac{1}{n^2}\right)} = 1.$$

Thus $E_n \subset K_n \cap H_n$.

C03 and T02: Limit of the lower bounds

By feasibility and G01,

$$\frac{w(K_n \cap H_n)}{w(K_n)} \geq \frac{\text{vol}_n(E_n)}{\kappa_n} = \frac{1}{2} \left(1 + \frac{1}{n-1}\right)^{(n-1)/2} \longrightarrow \frac{\sqrt{e}}{2}.$$

The limit follows from $(1 + 1/m)^m \rightarrow e$ and continuity of the square root. For every $r < r_*$, the lower bound exceeds r for all sufficiently large n , proving T02. It is unnecessary to prove that E_n is the maximum-volume ellipsoid in the retained side.

7. Additional conclusions

T03: Nonattainment in finite dimension. Under the assumptions of R01, every nondegenerate A satisfies $\det A < r_*$. If $\alpha \neq 1/2$, the relevant tangent in (S01.1) is strictly below C . If $\alpha = 1/2$ and $n \geq 2$, the case $j = 1$ of (R02.1) gives

$$\det A \leq \frac{1}{2} \left(1 + \frac{1}{4(n-1)}\right)^{2(n-1)} < \frac{\sqrt{e}}{2}.$$

If $n = 1$ and $\alpha = 1/2$, the determinant is $1/2$. A one-dimensional convex body is a nondegenerate closed interval, which is itself its maximum-volume inscribed ellipsoid. Cutting at its center produces two subintervals, each of half the original length. The optimal one-dimensional cutting ratio is therefore $1/2$.

T04: Constraint on the smallest semiaxis near equality. Under the same assumptions,

$$\log \frac{r_*}{\det A} \geq \begin{cases} \frac{1}{6} \log \frac{1}{2\alpha}, & \alpha \leq 1/2, \\ \frac{1}{42} \log(2\alpha), & \alpha \geq 1/2. \end{cases}$$

Subtract the appropriate tangent bound in (S01.1) from C , using $\log \det A \leq F_j(\log \alpha)$. Consequently, if $\varepsilon \geq 0$ and $\det A \geq r_* e^{-\varepsilon}$, then

$$\frac{1}{2} e^{-6\varepsilon} \leq \alpha \leq \frac{1}{2} e^{42\varepsilon}.$$

T05: Sequences of exact central cuts. Suppose K_0 is a convex body, $K_{k+1} = K_k \cap H_k$, and each H_k is a proper closed halfspace whose boundary passes through $c(K_k)$. Induction using T01 gives

$$w(K_N) \leq r_*^N w(K_0), \quad \log \frac{w(K_0)}{w(K_N)} \geq N(\log 2 - 1/2).$$

8. Alternative proof of M04

Under the assumptions of M04, inverse antitonicity and M01 imply, for $s > 0$,

$$\begin{aligned} b^\top (M + sI)^{-1} b &\geq u^\top A^2 \left(\frac{m}{\alpha^2} A^2 + sI \right)^{-1} u \\ &\geq \frac{\alpha^2}{m + s}. \end{aligned}$$

The last step uses the fact that $\lambda^2 / ((m/\alpha^2)\lambda^2 + s)$ is increasing for $\lambda > 0$.

Let $N = M + tbb^\top$. By (M02.1), interchange of trace with finite-dimensional integration, and integration by parts,

$$\mathrm{tr}(N^p) - \mathrm{tr}(M^p) = pc_p \int_0^\infty s^{p-1} \log(1 + t b^\top (M + sI)^{-1} b) ds.$$

More explicitly, let $L(s) = \log \det(N + sI) - \log \det(M + sI)$. Diagonalizing the fixed matrices M and N separately gives $L'(s) = \mathrm{tr}(N + sI)^{-1} - \mathrm{tr}(M + sI)^{-1}$. Taking the difference of traces yields $-c_p \int_0^\infty s^p L'(s) ds$, and integration by parts gives $pc_p \int_0^\infty s^{p-1} L(s) ds$.

For an invertible matrix Q , multilinearity of the determinant in its columns gives $\det(I + vw^\top) = 1 + w^\top v$: terms with two or more perturbed columns vanish because those columns are proportional, and the single-column terms sum to $w^\top v$. Taking $Q = M + sI$, $v = tQ^{-1}b$, and $w = b$, we obtain

$$\det(Q + tbb^\top) = \det Q (1 + t b^\top Q^{-1} b),$$

and hence

$$L(s) = \log(1 + t b^\top (M + sI)^{-1} b).$$

The boundary terms vanish: $L(s)$ has a finite limit as $s \downarrow 0$, and $L(s) = O(s^{-1})$ as $s \rightarrow \infty$, while $0 < p < 1$. The quadratic-form lower bound above and monotonicity of the logarithm therefore give

$$\mathrm{tr}(N^p) - \mathrm{tr}(M^p) \geq pc_p \int_0^\infty s^{p-1} \log\left(1 + \frac{t\alpha^2}{m+s}\right) ds = (m + t\alpha^2)^p - m^p,$$

where the last identity is the scalar case of the same integral identity. This proves (M04.1).